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Survey: Developing a Framework to Assess European Livestock Disease Outbreak Preparedness

This questionnaire aims to inform the development of a framework to assess European livestock disease outbreak preparedness regarding the following high-impact diseases: African horse sickness (AHS), African swine fever (ASF), Avian influenza (AI), Chronic wasting disease (CWD), and West Nile fever (WNF). This questionnaire is part of the Horizone Europe project “Ecology of Wild-life, Livestock, huMan and Infectious Diseases in changing environments” (WiLiMan-ID).

The data and knowledge produced through this project will provide guidance for national animal health authorities in Europe regarding the vulnerability of their livestock sectors to these five high priority livestock diseases, such as insights into the transmission pathways essential to the spread of these diseases within and between livestock populations and the economic impacts of these diseases on livestock sectors. Therefore, the data collected through this survey will aid in the formulation of economically sound, evidence-based animal health policy at both the farm and sector levels, complementing current knowledge of these diseases, and laying the foundation for effective contingency planning.

The data collected will not allow individual identification, as no personally identifiable information is requested. Unless you object in writing, the information collected through this survey may be published anonymously (i.e., scientific articles) and may be used in subsequent projects to refine or consolidate areas of research arising from this project.

The project consortium aims to make all data, aside from any GDPR-sensitive data and data obtained within the confines of a confidentiality agreement, findable, accessible, interoperable, and reusable (FAIR). The data management team at the Institut national de recherche pour l'agriculture, l'alimentation et l'environnement (INRAE) will ensure preservation of the data deposited at the project SharePoint for up to 5 years after the end of the project. If you are considering your participation in this survey, please carefully read and complete the subsequent “Survey Participation Ethics Consent Form” and respond to the questionnaire’s 17 questions.

Participation Ethics Consent Form

1. Survey and project information

In connection with your participation in a survey as part of a research project led by the University of Copenhagen, in collaboration with the Friedrich Loeffler Institute (FLI) and the INRAE, we hereby request your consent to us processing your questionnaire responses. We do so under the rules in the General Data Protection Regulation (GDPR).

Survey Title: Developing a Framework to Assess European Livestock Disease Outbreak Preparedness

Research Project Title: Ecology of Wild-life, Livestock, huMan and Infectious Diseases in changing environments (WiLiMan-ID)

Investigators: Philip Rasmussen; Beate Conrady

2. Type of data being collected

All questionnaire respondents will be anonymised and no personal data will be collected. All questions will be

directly related to your country of employment's livestock disease outbreak preparedness in terms of its **(i)** human resource holdings, **(ii)** training capacity, **(iii)** mobilisation capacity, and **(iv)** farm depopulation capacity in the event of a livestock disease outbreak emergency.

3. Implications of consent

Your consent to the processing of your questionnaire responses is voluntary and may be withdrawn until the time of anonymisation. You may change or withdraw your consent by contacting either Dr. Philip Rasmussen (phr@sund.ku.dk) or Dr. Beate Conrady (bcon@sund.ku.dk) from the University of Copenhagen.

4. Statement of consent

The participant confirms that they have read sections (1), (2), and (3) of this Survey Participation Ethics Consent Form, and that the information in these sections forms the basis on which they consent to the processing of their questionnaire responses by the WiLiMan-ID project. By completing the following questionnaire, the participant hereby gives their consent that the University of Copenhagen may process their questionnaire responses through the aforementioned research project.

Questionnaire (17 questions)

1. For which **country** are you responding?

2. Approximately **how many licensed veterinarians**, in total across all specialisations, are **currently working** in your country of employment?

Please enter the number of veterinarians.

3. Of the licensed veterinarians working in your country, what **percentage** have the following **specialisations**?

Please enter percentages across specialisations. Note that the total should sum to 100%.

Ruminants	
Horses	
Pigs	
Poultry	
Other and/or no specialisation	

4. How many private and public **institutions** offer accredited **veterinary education and/or training programmes** in your country of employment?

Please enter the number of private and public institutions.

Private institutions	
Public institutions	

5. Approximately how many veterinarians **graduated** from institutions in your country in the following years?

Please enter the number of graduates from private and public institutions across years.

	Private	Public
2023		
2022		
2021		

6. Approximately what percentage of veterinary graduates from within your country **stop practicing within your country** (i.e., change professions or leave the country to practice elsewhere) **after graduating**?

Please enter percentages across time periods.

Percentage that stop practicing **within 1 year** of graduating

Percentage that stop practicing **within 3 years** of graduating

Percentage that stop practicing **within 10 years** of graduating

7. In your country of employment, are **foreign veterinarians authorised** to operate in the event of a livestock disease **outbreak**?

Please select from the dropdown list.

Foreign veterinarians from EU countries

Foreign veterinarians from non-EU countries

8. In your country of employment, are there outbreak preparedness training programmes and/or exercises that are **mandatory** (i.e., required) for **veterinarians**?

Please select from the dropdown list.

9. In your country of employment, are there outbreak preparedness training programmes and/or exercises **available** to **livestock keepers/owners**?

Please select from the dropdown list.

10. Across the livestock diseases listed below, please indicate the **frequency of emergency outbreak training exercises** in your country of employment.

Please select a frequency from the dropdown list for each disease.

African horse sickness (AHS)

African swine fever (ASF)

Avian influenza (AI)

Chronic wasting disease (CWD)

West Nile fever (WNF)

11. For the veterinary resources listed below, is your country part of a formal international **resource-sharing agreement** in the event of an outbreak emergency?

If your country is part of a formal resource-sharing agreement, please describe the agreement and list the country, countries, or group of countries (e.g., Schengen area, EU, etc.) that are part of the agreement and any other relevant aspects (e.g., disease-specific aspects of the agreement, species-specific aspects, time periods, etc.).

If your country is not part of a formal resource-sharing agreement, please enter "Not applicable".

Human resources

Vehicles, including rendering trucks	
Medical supplies	
Laboratory access	
Vaccines	

12. This question is about your country's **vehicle capacity**. Across the species listed below, approximately **how** many vehicles designated for the transport of **live animals** and **animal carcasses** are available in your country of employment?

Please enter the number of vehicles designated for the transport of live animals and animal carcasses across each species.

	Vehicles (live animals)	Vehicles (animal carcasses)
Cattle		
Horses		
Pigs		
Poultry		
Small ruminants		

13. This question is about your country's **depopulation capacity.** Across the species listed below, approximately how many animals could be **depopulated per day** across your country? Please consider your country's depopulation capacity both in normal circumstances (i.e., the **average capacity** that would be sustainable on an everyday basis) and in emergency circumstances (i.e., the **maximum capacity** that would only be sustainable for a short period during an outbreak).

Please enter the number of animals per day, in normal and in emergency circumstances, across species.

	Animals per day (normal)	Animals per day (emergency)
Cattle		
Horses		
Pigs		
Poultry		
Small ruminants		

14. This question is about your country's **laboratory capacity.** Across the diseases and test types listed below, approximately how many samples **per week could be tested** across your country? Please consider your country's laboratory capacity both in normal circumstances (i.e., the **average capacity** that would be sustainable on an everyday basis) and in emergency circumstances (i.e., the **maximum capacity** that would only be sustainable for a short period during an outbreak).

Please enter the number of samples per week, in normal and in emergency circumstances, across diseases and test types.

	Samples per week (normal)	Samples per week (emergency)
African horse sickness (AHS)		
Polymerase chain reaction (PCR)		

Enzyme-linked immunosorbent assay (ELISA)		
African swine fever (ASF)		
PCR		
ELISA		
Avian influenza (AI)		
PCR		
ELISA		
Chronic wasting disease (CWD)		
PCR		
ELISA		
West Nile fever (WNF)		
PCR		
ELISA		

15. For the diseases listed below, is your country **doing more to manage the risk of introduction** than the **minimum required** by EU law?

If your country is doing more to manage the risk of introduction, please briefly describe the additional measures being implemented in your country. If your country is not doing more to manage the risk of introduction, please enter "Not applicable".

African horse sickness (AHS)	
African swine fever (ASF)	
Avian influenza (AI)	
Chronic wasting disease (CWD)	
West Nile fever (WNF)	

16. Does your country **collect data** on the **population** and/or the **disease status** (e.g., Chronic wasting disease, African swine fever, or other diseases) of the following **wildlife species**?

If your country is collecting data, please briefly describe the type of data being collected. If your country is not collecting data, please enter "Not applicable".

	Population data	Disease status data
Deer		
Wild boar		

17. Does your country have **monitoring programmes** for the following **ectoparasites**?

If your country has a monitoring programme, please briefly describe the programme, and if applicable, the type of data being collected. If your country has no monitoring programme, please enter "Not applicable".

	Monitoring programme
Mosquitos	

Ticks

Thank you for your participation!

Please do not hesitate to contact either Philip Rasmussen (phr@sund.ku.dk) or Beate Conrady (bcon@sund.ku.dk) from the University of Copenhagen if you require any additional information or if you have any comments or questions.